2025 CS2 Big Event Team Analysis

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0.1 Project Overview

This report compares the 2025 Big Event map-level performance of five CS2 teams: Vitality, Falcons, Spirit, FURIA, and MOUZ. The main focus is map win rate, map-pool coverage, and an overall descriptive rating.

0.2 CS2 Performance Metrics

This project uses several CS2 map-level metrics to compare team performance. CS2, or Counter-Strike 2, is a competitive first-person shooter where professional teams play on different maps. In this report, each map is treated as one observation, so the analysis focuses on map-level performance rather than only full match results.

The definitions in Table 1 explain the main terms used in the tables and figures.

Table 1: CS2 performance metrics used in this report

Metric	Meaning
Map	The CS2 map played in each record, such as Ancient, Nuke, Mirage, Inferno, or Anubis.
Total Maps	The total number of maps played by a team. A larger sample size makes the result more reliable.
Wins	The number of maps a team won. This counts individual maps, not full BO3 or BO5 matches.
Losses	The number of maps a team lost.
Win Rate	The percentage of maps won by a team, calculated as `Wins / Total Maps`.
Map-specific Win Rate	The win rate of a team on a specific map. This helps identify strong maps and weak maps.
Target Score	The number of rounds won by the target team on a map.
Opponent Score	The number of rounds won by the opponent on the same map.
Round Difference	The difference between the team's rounds and the opponent's rounds, calculated as `Target Score - Opponent Score`.
Average Round Difference	The average round difference across maps. A higher value suggests stronger map control and dominance.
Maps Played	The number of different maps a team has played. This reflects the width of the team's map pool.
Map Pool Coverage	The percentage of the full map pool covered by a team. It shows whether a team has a wide or narrow map pool.
Overall Score	A custom final rating used in this project. It combines win rate, map pool coverage, and sample size.

In this report, **map win rate** is the most important metric because it directly measures how often a team wins individual maps. **Average round difference** adds more detail by showing whether a team wins maps comfortably or only by close margins. **Map pool coverage** is also important because a stronger team should usually perform well across multiple maps instead of relying on only one or two strong maps.

0.3 Map Win Rate Comparison

The radar charts compare each team's map win rate across the active map pool. A wider and more balanced shape suggests stronger map-pool coverage. Figure 1 gives an overall comparison, while Figure 2 separates the teams for easier reading.

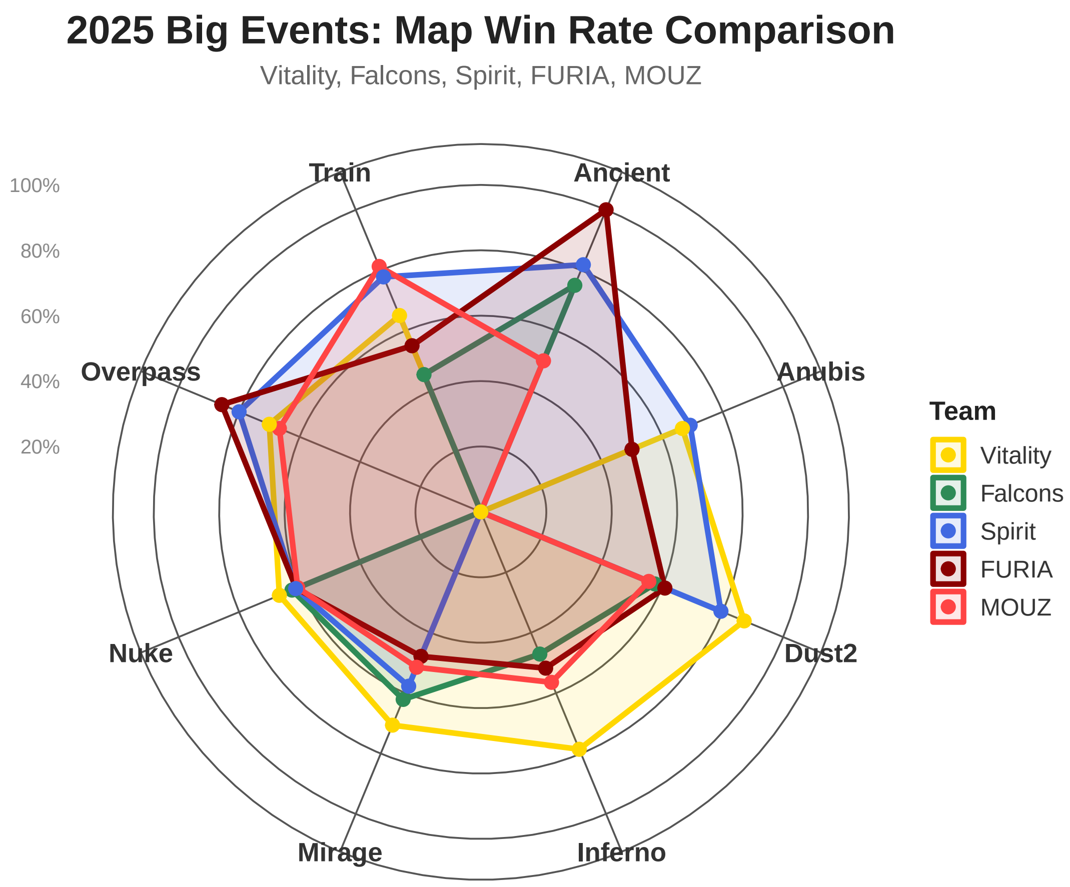

Figure 1: Overlay radar chart of map win rate for the five selected teams.

Figure 1 gives a compact view of the full comparison. Teams with larger shapes have stronger results across more maps, while uneven shapes suggest that a team is more dependent on specific maps.

Figure 2 makes the map-level patterns easier to compare because each team is shown in its own panel.

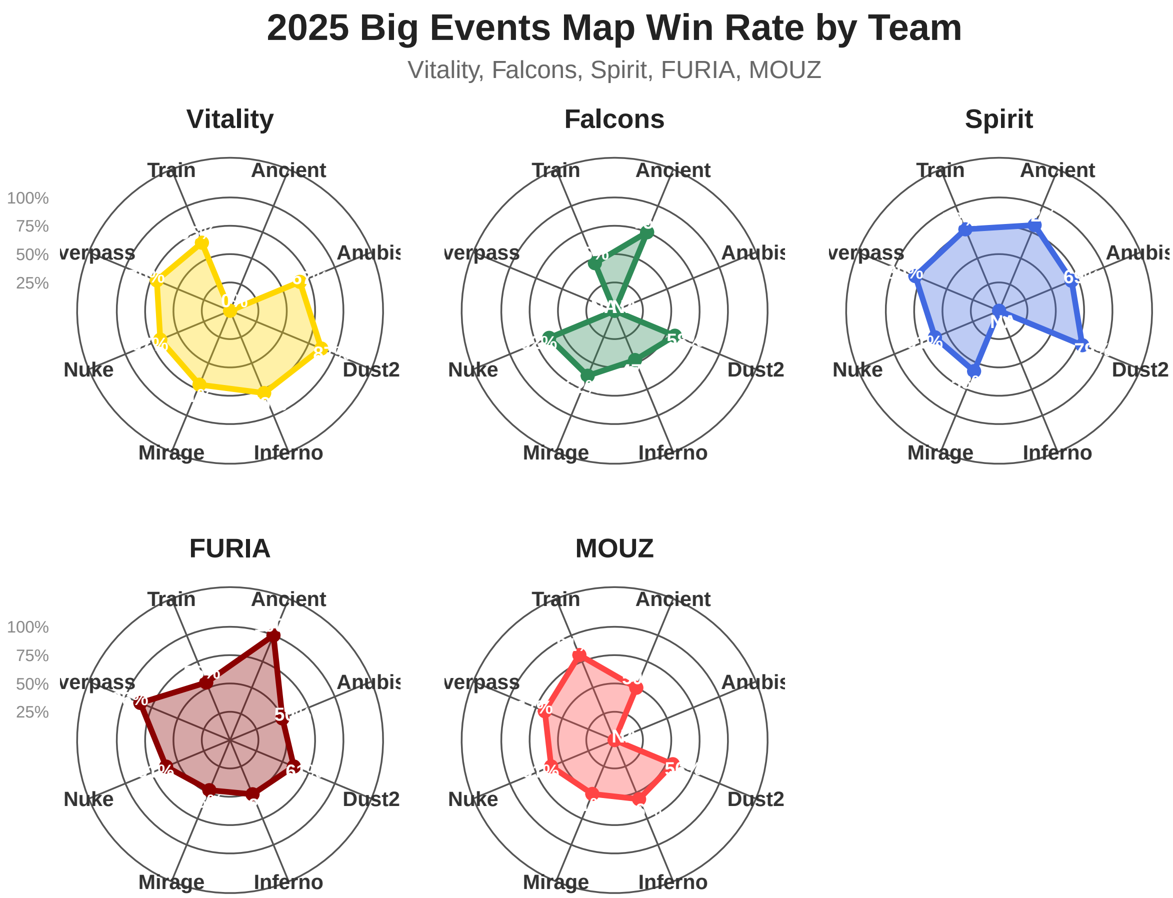

Figure 2: Separate radar charts of map win rate by team.

0.4 Map Pool Performance

The heatmap in Figure 3 shows each team's win rate on each map. Each label shows the win rate and the number of maps played. Cells marked as "No maps" mean that the team did not play that map in this dataset.

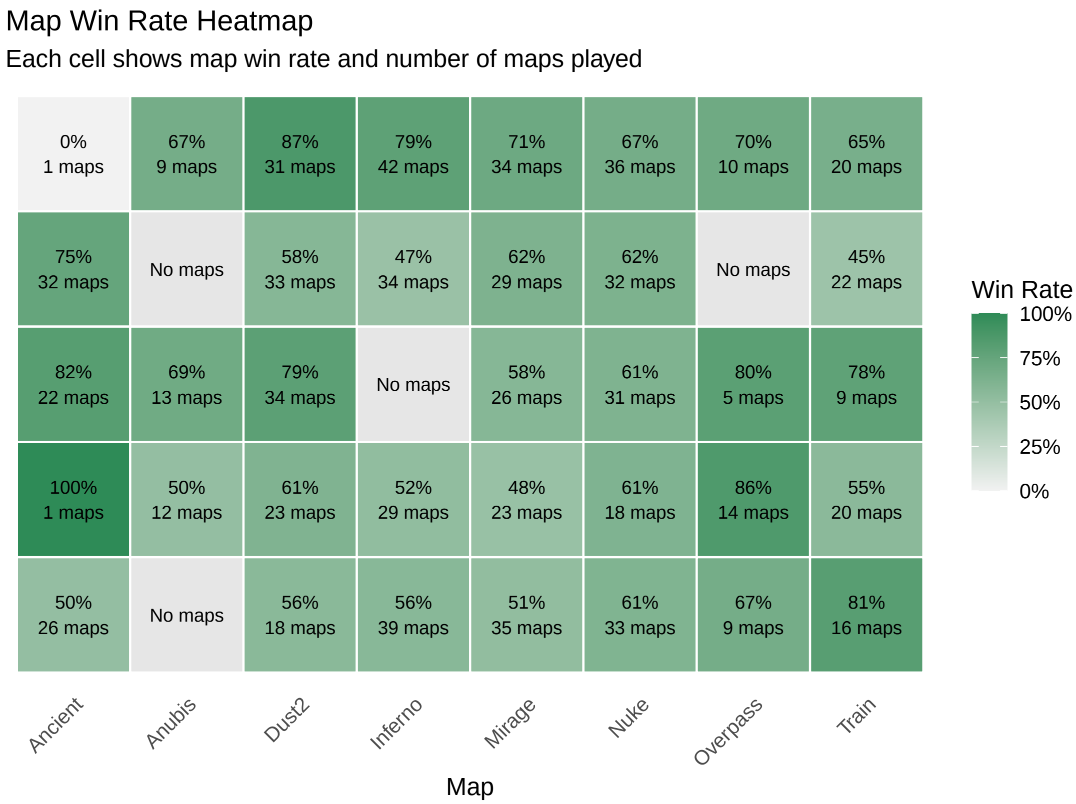

Figure 3: Map win rate heatmap with the number of maps played.

Figure 3 is useful because it shows both performance and sample size in the same place. A high win rate based on only one map should be read more carefully than a high win rate across many maps.

0.5 Overall Team Rating

The overall rating gives the most weight to direct winning performance, while still rewarding teams that played across a wider map pool and had a larger sample size. Table 2 lists the rating components, and Figure 4 ranks the teams from highest to lowest.

Table 2: Overall team summary and rating.

Target-Team	TotalMaps	Wins	Losses	WinRate	AvgRoundDifference	MapPoolCoverage	OverallScore
Vitality	183	134	49	73.2%	2.97	100.0%	81.26
Spirit	140	99	41	70.7%	2.61	87.5%	74.65
FURIA	140	81	59	57.9%	1.07	100.0%	68.15
MOUZ	176	102	74	58.0%	0.73	87.5%	67.69
Falcons	182	107	75	58.8%	1.29	75.0%	66.10

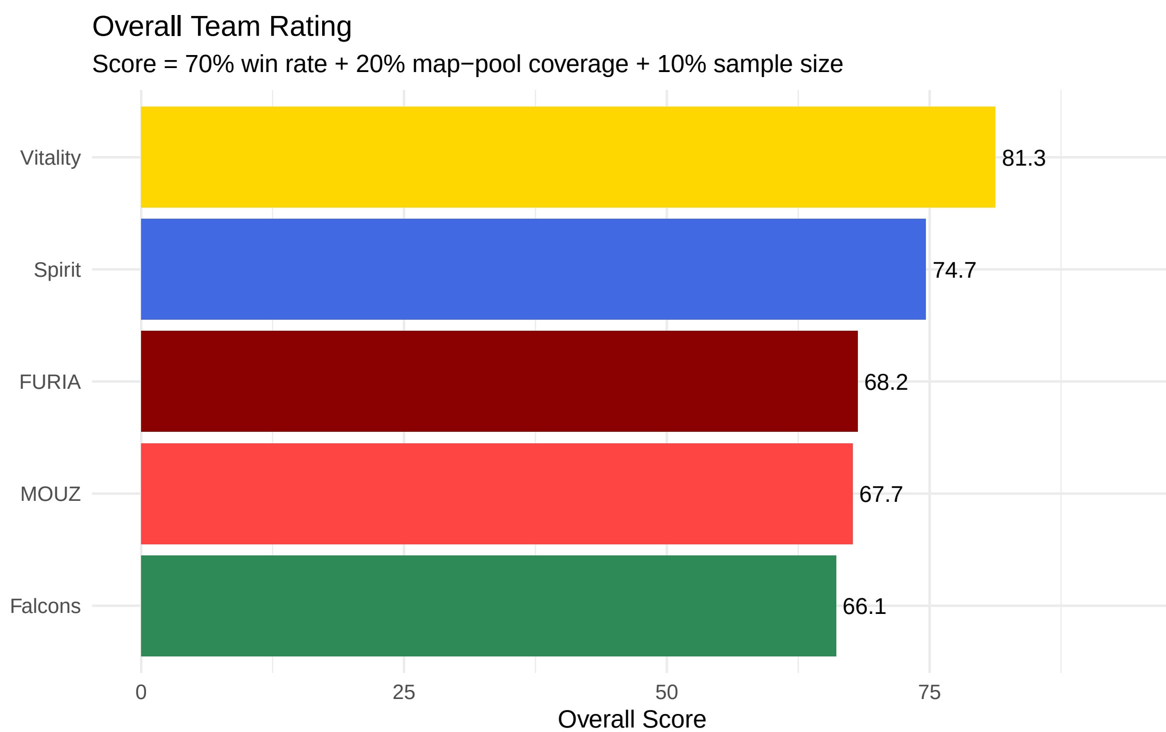

Figure 4: Overall team rating ranked from highest to lowest.

The ranking in Figure 4 summarizes the data story: a team needs both strong win rate and enough map-pool depth to finish near the top. The table gives the exact values behind the chart, while the radar charts and heatmap explain where those differences come from.

0.6 Written Summary

Based on the rating formula, Vitality ranks first overall. This means the team combines strong win rate, map-pool coverage, and enough map sample size better than the other selected teams in this dataset.

The radar charts provide a more detailed map-level view. They show that the teams do not perform equally across every map, so a team's overall score should be read together with its map-specific strengths and weaknesses.

0.7 Limitations

This rating is a descriptive score, not a predictive model. It summarizes the selected HLTV Big Event data, but it does not fully measure opponent strength, veto strategy, roster changes, or tournament context.

0.8 Reference

HLTV.org. (n.d.). Counter-Strike statistics database. Retrieved May 5, 2026, from HLTV.org.

1 Code Appendix

The code below was used to clean the data and create the tables and figures in this report.

```
library(tidyverse)
library(scales)

hltv <- read_csv("hltv_2025_bigevents_matches_all.csv")

target_teams <- c("Vitality", "Spirit", "Falcons", "FURIA", "MOUZ")

target_matches <- hltv %>%
  filter(Team1 %in% target_teams | Team2 %in% target_teams)

team1_view <- target_matches %>%
  filter(Team1 %in% target_teams) %>%
  transmute(
    Date,
    TargetTeam = Team1,
    Opponent = Team2,
    TargetScore = Team1Score,
    OpponentScore = Team2Score,
    Map,
    Event,
    MatchURL
  )

team2_view <- target_matches %>%
  filter(Team2 %in% target_teams) %>%
  transmute(
    Date,
    TargetTeam = Team2,
    Opponent = Team1,
    TargetScore = Team2Score,
    OpponentScore = Team1Score,
    Map,
    Event,
    MatchURL
  )

target_team_maps <- bind_rows(team1_view, team2_view) %>%
  mutate(
```

```r
    Result = case_when(
      TargetScore > OpponentScore ~ "Win",
      TargetScore < OpponentScore ~ "Loss",
      TRUE ~ "Draw"
    )
  ) %>%
  arrange(TargetTeam, Date, Map)

write_csv(target_matches, "hltv_target_matches_raw.csv")
write_csv(target_team_maps, "hltv_target_team_maps_clean.csv")

map_order  <- c("Ancient", "Anubis", "Dust2", "Inferno",
                "Mirage", "Nuke", "Overpass", "Train")

map_winrate <- target_team_maps %>%
  group_by(TargetTeam, Map) %>%
  summarise(
    TotalMaps = n(),
    Wins = sum(Result == "Win"),
    Losses = sum(Result == "Loss"),
    WinRate = Wins / TotalMaps,
    .groups = "drop"
  ) %>%
  complete(
    TargetTeam = target_teams,
    Map = map_order,
    fill = list(TotalMaps = 0, Wins = 0, Losses = 0)
  ) %>%
  mutate(
    WinRate = if_else(TotalMaps == 0, NA_real_, WinRate),
    WinRatePlot = replace_na(WinRate, 0)
  )

write_csv(map_winrate, "hltv2025.csv")

map_winrate <- read_csv("hltv2025.csv")

team_order <- c("Vitality", "Falcons", "Spirit", "FURIA", "MOUZ")

team_colors <- c(
  "Vitality" = "#FFD700",
  "Falcons"  = "#2E8B57",
```

```
  "Spirit"   = "#4169E1",
  "FURIA"    = "#8B0000",
  "MOUZ"     = "#FF4444"
)

coord_radar <- function(theta = "x", start = 0, direction = 1) {
  theta <- match.arg(theta, c("x", "y"))
  r <- if (theta == "x") "y" else "x"
  ggproto("CoordRadar", CoordPolar,
          theta = theta, r = r, start = start,
          direction = sign(direction),
          is_linear = function() TRUE)
}

plot_data <- map_winrate %>%
  mutate(
    Map        = factor(Map, levels = map_order),
    TargetTeam = factor(TargetTeam, levels = team_order),
    label_text = ifelse(is.na(WinRate), "NA",
                        paste0(round(WinRatePlot * 100), "%"))
  )

p1 <- ggplot(plot_data,
             aes(x = Map, y = WinRatePlot,
                 group = TargetTeam, color = TargetTeam, fill = TargetTeam)) +
  geom_polygon(alpha = 0.12, linewidth = 1.2) +
  geom_point(size = 2.5) +
  coord_radar() +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = c(0.2, 0.4, 0.6, 0.8, 1.0),
    labels = percent_format(accuracy = 1)
  ) +
  scale_color_manual(values = team_colors) +
  scale_fill_manual(values = team_colors) +
  labs(
    title    = "2025 Big Events: Map Win Rate Comparison",
    subtitle = "Vitality, Falcons, Spirit, FURIA, MOUZ",
    color    = "Team",
    fill     = "Team",
    x = NULL, y = NULL
  ) +
```

```r
  theme_minimal(base_size = 12)

p2 <- ggplot(plot_data,
             aes(x = Map, y = WinRatePlot,
                 group = TargetTeam, color = TargetTeam, fill = TargetTeam)) +
  geom_polygon(alpha = 0.35, linewidth = 1.3) +
  geom_point(size = 2.5) +
  geom_text(
    aes(label = label_text, y = WinRatePlot + 0.10),
    color = "white", size = 3, fontface = "bold"
  ) +
  coord_radar() +
  scale_y_continuous(
    limits = c(0, 1.2),
    breaks = c(0.25, 0.5, 0.75, 1.0),
    labels = percent_format(accuracy = 1)
  ) +
  scale_color_manual(values = team_colors) +
  scale_fill_manual(values = team_colors) +
  facet_wrap(~ TargetTeam, ncol = 3) +
  labs(
    title    = "2025 Big Events Map Win Rate by Team",
    subtitle = "Vitality, Falcons, Spirit, FURIA, MOUZ",
    x = NULL, y = NULL
  ) +
  theme_minimal(base_size = 12)

heatmap_data <- map_winrate %>%
  mutate(
    Map = factor(Map, levels = map_order),
    TargetTeam = factor(TargetTeam, levels = rev(team_order)),
    HeatmapWinRate = if_else(TotalMaps == 0, NA_real_, WinRate),
    label_text = if_else(
      TotalMaps == 0,
      "No maps",
      paste0(round(WinRate * 100), "%\n", TotalMaps, " maps")
    )
  )

ggplot(heatmap_data, aes(x = Map, y = TargetTeam, fill = HeatmapWinRate)) +
  geom_tile(color = "white", linewidth = 0.5) +
  geom_text(aes(label = label_text), size = 3) +
```

```r
  scale_fill_gradient(
    low = "#F2F2F2",
    high = "#2E8B57",
    limits = c(0, 1),
    labels = scales::percent_format(accuracy = 1),
    na.value = "#E6E6E6",
    name = "Win Rate"
  ) +
  labs(
    title = "Map Win Rate Heatmap",
    subtitle = "Each cell shows map win rate and number of maps played",
    x = "Map",
    y = NULL
  ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

team_summary <- target_team_maps %>%
  group_by(TargetTeam) %>%
  summarise(
    TotalMaps = n(),
    Wins = sum(Result == "Win"),
    Losses = sum(Result == "Loss"),
    WinRate = Wins / TotalMaps,
    AvgRoundDifference = mean(TargetScore - OpponentScore),
    MapPoolCoverage = n_distinct(Map) / length(map_order),
    .groups = "drop"
  ) %>%
  mutate(
    SampleSizeScore = TotalMaps / max(TotalMaps),
    OverallScore = 100 * (
      0.70 * WinRate +
      0.20 * MapPoolCoverage +
      0.10 * SampleSizeScore
    )
  ) %>%
  arrange(desc(OverallScore))

team_summary %>%
```

```
  mutate(
    WinRate = scales::percent(WinRate, accuracy = 0.1),
    MapPoolCoverage = scales::percent(MapPoolCoverage, accuracy = 0.1),
    AvgRoundDifference = round(AvgRoundDifference, 2),
    OverallScore = round(OverallScore, 2)
  ) %>%
  select(
    TargetTeam, TotalMaps, Wins, Losses, WinRate,
    AvgRoundDifference, MapPoolCoverage, OverallScore
  ) %>%
  knitr::kable()

ggplot(team_summary,
       aes(x = reorder(TargetTeam, OverallScore),
           y = OverallScore,
           fill = TargetTeam)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = round(OverallScore, 1)), hjust = -0.15, size = 4) +
  coord_flip() +
  scale_fill_manual(values = team_colors) +
  labs(
    title = "Overall Team Rating",
    subtitle = "Score = 70% win rate + 20% map-pool coverage + 10% sample size",
    x = NULL,
    y = "Overall Score"
  ) +
  ylim(0, max(team_summary$OverallScore) * 1.15) +
  theme_minimal(base_size = 12)
```